Human Computer Interaction

Module Name:	Human Computer Interaction
Module Code:	HCIN6222
Student Name & Surname:	Siseko Landingwe
Student Number:	ST10482760
Submission Date:	22 October 2026

1.

1.1 Review of Lecturer Feedback

Following the submission of Part 1, feedback was provided regarding the initial planning of the PeerConnect website. The main areas noted for improvement included: clarifying how students would be prevented from accidentally deleting group assignment notes, making the navigation less crowded on mobile screens, and providing clearer visual cues for accessibility and error recovery.

1.2 Summary of Adjustments Made to Planning

To address this feedback before building the interactive prototype, I adjusted my site design and layout plans as follows:

- **Safety & Error Handling:** I redesigned the delete action so that clicking 'Remove Note' opens a modal popup asking 'Are you sure you want to delete this resource?', along with an instant 'Undo' toast notification at the bottom of the screen (Norman, 2013).
- **Interface Layout & Navigation:** Instead of keeping all navigation options permanently visible on top, I planned a collapsible top navigation bar and a simplified side drawer for smaller displays to reduce clutter (Shneiderman et al., 2016).
- **Visual Contrast & Accessibility:** I adjusted the color palette to ensure all body text meets WCAG AAA contrast ratios (at least 7:1) against background cards, ensuring comfortable reading for visually impaired students (World Wide Web Consortium, 2008.
- **Social & Gamification Cues:** I added dedicated UI sections for live presence dots (showing who is currently active in a study room) and peer-review endorsement badges to make collaboration feel more social and rewarding (Preece, Rogers and Sharp, 2015).

2. WEBSITE OVERVIEW AND PROTOTYPE ARCHITECTURE

2.1 Refined Platform Overview

PeerConnect is an interactive web platform designed specifically for tertiary students who need a focused workspace for academic group tasks, peer study matching, and module resource sharing (Sharp, Rogers and Preece, 2019). Unlike generic chat apps, PeerConnect organizes student interactions around module codes and specific assignment goals.

2.2 Core Prototype Features & Flow

The developed prototype implements four main functional sections:

1. Peer Matching & Study Group Formation: Allows students to search for classmates by module code (e.g., HCIN6222) and filter by study availability or specific topics like UI design or database queries (Shneiderman et al., 2016).

2. Group Task Workspace: A Kanban task board featuring drag-and-drop cards across 'To Do', 'In Progress', and 'Completed' columns, helping group members track assignment progress in real time (Norman, 2013).

3. Resource Exchange & Micro-Tutoring Hub: A repository where students can upload study summaries and code snippets, backed by a star-rating system and peer endorsement badges (Dix et al., 2004).

4. Module Q&A Discussions: Categorized message threads where students can ask lecture-related questions and upvote helpful answers from peers (Preece, Rogers and Sharp, 2015).

3. PROTOTYPE IMPLEMENTATION OF HCI ELEMENTS

3.1 Usability Goals

Usability goals evaluate how effectively and efficiently a user can complete tasks using the interface (Sharp, Rogers and Preece, 2019:18). Below is how each usability goal is visually represented in the prototype:

1. Effectiveness: The prototype guides students directly from their home dashboard to join a study group or find notes without getting lost in unnecessary submenus (Dix et al., 2004).

2. Efficiency: Core tasks like searching for a peer tutor or creating a new group task take three clicks or fewer. Shortcut buttons are placed directly on the main dashboard header (Shneiderman et al., 2016).

3. Safety: To prevent accidental data loss, clicking a delete icon triggers a confirmation dialog asking the user to verify the action, accompanied by an instant 'Undo' bar (Norman, 2013).

4. Utility: The platform includes all tools required for academic teamwork, including live task boards, shared file uploads, meeting scheduling, and chat threads (Sharp, Rogers and Preece, 2019).

5. Learnability: The interface uses standard web layouts, clean labels, and familiar icon conventions so first-time users can understand how to navigate the site right away (Preece, Rogers and Sharp, 2015).

6. Memorability: Clear visual icons (e.g., search glass for lookup, folder for resources) and consistent top navigation mean returning students remember how to use the site easily (Shneiderman et al., 2016).

3.2 Desirable Aspects of User Experience (UX)

User Experience (UX) focuses on how satisfying and engaging the platform feels during use (Sharp, Rogers and Preece, 2019):

- **Satisfying and Rewarding:** Moving a task card to 'Completed' displays a green progress bar update and celebratory micro-animation, giving students a sense of accomplishment (Nielsen, 1994).
- **Engaging and Motivating:** Peer badges (e.g., 'Top HCI Tutor' or 'Helpful Note Author') reward active participation and keep students motivated (Preece, Rogers and Sharp, 2015).

• **Helpful and Empowering:** Automated study match suggestions and organized assignment checklists help students feel supported and confident in managing their workload (Shneiderman et al., 2016).

• **Enjoyable and Stress-Free:** The uncluttered card layout, generous spacing, and soothing blue color scheme create a calm interface that reduces assignment stress (Norman, 2013).

3.3 Design Principles

Design principles guide the placement and styling of UI components to ensure intuitive operation (Norman, 2013):

1. Visibility: Primary action buttons like 'Create Study Group' and 'Upload Notes' are highlighted in dark blue at the top of the interface so they are immediately obvious (Norman, 2013).

2. Feedback: When a user saves a new task or uploads a document, a banner pops up confirming 'Resource successfully saved!', providing immediate feedback (Sharp, Rogers and Preece, 2019).

3. Constraints: Form submit buttons remain grayed-out until all required text fields (like module name and topic) are properly filled in, preventing incomplete submissions (Norman, 2013).

4. Consistency: All pages maintain consistent fonts, card corner radiuses, and button styling (primary buttons are always solid dark blue, secondary buttons are outlined) (Dix et al., 2004).

5. Affordance: Buttons feature slight shadow gradients and hover hover-effects to show they can be clicked, while hyperlinked text is visibly underlined (Norman, 2013).

6. Mapping: The Kanban board flows logically from left to right ('To Do' -> 'In Progress' -> 'Completed'), matching how users naturally track task progression (Shneiderman et al., 2016).

3.4 Interaction Types

Interaction types describe how users control and navigate the platform (Sharp, Rogers and Preece, 2019:50):

• **Instructing:** Students fill out search boxes and click filter checkboxes to instruct the site to show specific study groups (Dix et al., 2004).

• **Conversational:** A built-in study assistant chatbot ('PeerBot') allows students to ask questions like 'When is my next HCI meeting?' in plain text (Preece, Rogers and Sharp, 2015).

• **Manipulating:** Students drag and drop assignment task cards directly across columns on the group workspace board (Shneiderman et al., 2016).

• **Exploring:** Students can click through different module tabs and study topic cards to explore shared notes and discussion threads (Sharp, Rogers and Preece, 2019).

3.5 Social Interactions

PeerConnect includes targeted social features to make group work smooth and collaborative (Preece, Rogers and Sharp, 2015):

• **Synchronous Co-Presence:** Green status dots show which study group members are currently online, and live typing indicators show when someone is adding notes in real time (Dix et al., 2004).

• **Asynchronous Communication:** Discussion threads and comment boxes allow team members to leave messages and feedback even if classmates log in at different times (Sharp, Rogers and Preece, 2019).

• **Social Accountability:** Task assignment tags display student avatars next to their assigned tasks on the Kanban board, keeping everyone accountable for their portion of work (Preece, Rogers and Sharp, 2015).

3.6 Emotional Interaction

Emotional interaction examines how the interface affects student mood and confidence (Norman, 2004:35). Grounded in Norman's Emotional Design Framework:

1. Visceral Level (Appearance): Soft navy blue and white background colors, neat margins, and clean typography make the site look inviting and reduce academic stress (Norman, 2004).

2. Behavioral Level (Usability): Smooth card animations and fast response times make students feel competent and in full control of their study tasks (Norman, 2004).

3. Reflective Level (Meaning): A personal dashboard displays completed study goals and peer endorsements, helping students feel proud of their study progress (Sharp, Rogers and Preece, 2019).

3.7 Web Content Accessibility Guidelines (WCAG) 2.0 Compliance

The prototype incorporates WCAG 2.0 principles to ensure equal access for students with disabilities (World Wide Web Consortium, 2008):

• **Perceivable:** High-contrast text styling (dark text on white/light gray backgrounds) ensures easy readability. All icons include alternative text attributes for screen readers (World Wide Web Consortium, 2008).

• **Operable:** All buttons, links, and forms can be focused and triggered using only the keyboard (Tab and Enter keys), highlighted by clear visual focus borders (World Wide Web Consortium, 2008).

• **Understandable:** Input fields feature explicit labels and friendly error messages (e.g., 'Please enter a valid module code') to guide students through form entry (World Wide Web Consortium, 2008).

• **Robust:** The HTML code follows standard semantic structures (such as `<header>`, `<nav>`, `<main>`, and `<button>`), ensuring screen readers interpret the site properly (World Wide Web Consortium, 2008).

4. REFERENCE LIST

Dix, A., Finlay, J., Abowd, G.D. and Beale, R., 2004. Human-Computer Interaction. 3rd ed. Harlow: Pearson Education.

Nielsen, J., 1994. Usability Engineering. San Diego: Academic Press.

Norman, D.A., 2004. Emotional Design: Why We Love (or Hate) Everyday Things. New York: Basic Books.

Norman, D.A., 2013. The Design of Everyday Things. Revised and expanded ed. New York: Basic Books.

Preece, J., Rogers, Y. and Sharp, H., 2015. Interaction Design: Beyond Human-Computer Interaction. 4th ed. Chichester: John Wiley & Sons.

Sharp, H., Rogers, Y. and Preece, J., 2019. Interaction Design: Beyond Human-Computer Interaction. 5th ed. Indianapolis: John Wiley & Sons.

Shneiderman, B., Plaisant, C., Cohen, M., Jacobs, S., Elmqvist, N. and Diakopoulos, N., 2016. Designing the User Interface: Strategies for Effective Human-Computer Interaction. 6th ed. Boston: Pearson.

World Wide Web Consortium, 2008. Web Content Accessibility Guidelines (WCAG) 2.0. W3C Recommendation. Available at: <https://www.w3.org/TR/WCAG20/>